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Exploring the relationship between basketball rotation and competitive performance using substitution network analysis

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ABSTRACT

The aim of this exploratory study is (1) to determine the relationship between substitution network (Sub-N) parameters and teams' standings and (2) to find out the

and explore the association between players' percentages and team's standing within the obtained substitution groups. A total of 574,214 substitution events during the last 10 NBA seasons were analysed to construct Sub-N for each team observation. Three different player groups were obtained after clustering their playing time, clustering coefficient and vulnerability. Team's clustering coefficient, standard deviation of vulnerability and out-degree centrality of starters exhibited moderate to strong correlations with team's standing during playoffs (r = 0.54–0.76). The regression models showed that defensive win share (beta = 0.54–0.67), turnovers (−0.15 to −0.25) and assists (0.12–0.26) were predictive for all players' net ratings, and the role players who scored more points presented higher net ratings (0.34). Finally, players from top-playoff teams exhibited lower absolute value of vulnerabilities (r = 0.80). The findings demonstrate the feasibility of Sub-N for exploring the association between rotation and competitive performance, and provide quantitative reference for coaching staff to optimize substitution structures and rosters.

KEYWORDS:

Basketball rotation social network performance evaluation

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